MIL-C-646A(MU)
13 September 1968
SUPERSEDING
JAN-C-646
5 August 1948

MILITARY SPECIFICATION

CUPS, CARTRIDGE-CASE, STEEL (FOR SMALL ARMS AMMUNITION)

1. SCOPE

1.1 This specification covers annealed steel cups used in the manufacture of cartridge cases for small arms ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1232	Visual Inspection Standard for Cups and Disks used in Small Arms Ammunition Manufacture

FSC 1305

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2.1 (Continued)

DRAWINGS

US Army Munitions Command

Any cartridge case cup drawing specified in the contract or purchase order (see 6.1).

D7553671	Box, Wirebound, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly
F7553844	Box, Shipping for Small Arms Ammunition Cups and Disks for Bullet Jackets and Cases - List of Parts and Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge case cup shall comply with the requirements specified on the applicable drawings and referenced specifications, and shall process satisfactorily into a cartridge case with metal defects not to exceed 0.10 percent.

3.2 Hardness.- The average hardness of any cup shall not exceed Rockwell B65 and no individual reading shall exceed Rockwell B70, unless otherwise indicated.

3.3 Condition.- The cups shall be furnished spheroidized annealed. The carbides in the cups shall be spheroidized and uniformly distributed. Lamellar pearlite shall not be present to an extent which will interfere with the purpose for which the cups are intended. The cups shall be protected by an adequate and readily removable rust preventative to insure their receipt in satisfactory condition.

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3.4 Workmanship.- The requirements for workmanship are as specified in the applicable drawings, referenced specifications and the following:

3.4.1 Metal defects.- The cup shall be free of laminations, slivers, laps, cracks, scale, scratches, dents, bent sidewalls, V'd edges and wire edges.

3.4.2 Foreign matter.- The cup shall be clean and free of oil, grease, dirt, grit, oxidation, stains and chips.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5200 cups. The samples shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination-and test.- After inspection and provisional acceptance at source, 200 cups shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Workability test.- The remaining 5000 cups shall be used for performance of a work test.

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.- A lot shall consist only of material from one heat of steel, of the same size cups, of the same condition.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of cups shall be identified as to type, caliber and model, as well as with a lot number and the manufacturer's identification as assigned by the procuring activity.

4.3.2 Examination.- Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The acceptable quality level (AQL) for the major class shall be 0.25 percent. All non-conforming cartridge cups shall be rejected.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

TABLE I

Defect	Method of Inspection
MAJOR	
101. Scaly metal, slivers, wire edges	Visual
102. Improperly formed or mutilated	Visual
103. Laminations or laps 1/	Visual
104. Outside dimeter	Gage
105. Base thickness	Gage
106. Wall thickness	Gage
107. Wall thickness variation	Gage
108. Wall height variation	Gage
109. Weight	Scale
110. Scratches	Visual
111. Seams and cracks 1/	Visual
112. Oil, grease, dirt	Visual
113. Dents or bent edges	Visual
114. Oxidation	Visual

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4.3.2.1 (Cont'd)

1/ Cups found in the visual inspection with these defects shall be sectioned, polished, etched and examined microscopically.

NOTE: Refer to applicable steel case cup section of MIL-STD-1232 for visual defect standards for visual defects listed in Table I.

4.3.3 Tests.- The following tests shall be conducted in accordance with methods and procedures specified in 4.4.

4,3.3.1 Test sample.- The quantities for the various tests shall be as specified in Table II. Only cups having met the visual, dimensional and weight requirements shall be used in the tests, and shall have been selected in such manner that the sample is representative of the entire lot.

TABLE II

Tests	Number of Cups	Requirement Paragraph
Hardness 1/	25	3.2
Micro-examination 2/	3	3.3
Laminations 1/ 3/	25	3.4.1
Seams and cracks 1/ 3/	25	3.4.1

1/ Cups used for hardness test shall also be used in the examination for laminations, seams and cracks. Failure of any cup to comply with the applicable requirement shall be cause for rejection of the lot.

2/ Cups shall be suitably sectioned, polished, etched and examined for compliance with the standard for microstructure at a magnification of 1000X.

3/ Cups shall resectioned, polished, etched and examined with a microscope. Cups found in the visual inspection with these defects shall also be sectioned by this method. If a break extends more than 0.020 inch below the lowest point of the mouth level, the cup shall be called laminated. If seams and cracks extend into the body of the metal, the cups shall be considered defective.

4.3.3.2 Workability.- A sample of 5000 cups from the initial production sample shall be subjected to a work test to determine compliance with the processing requirement specified in 3.1 and 6.1.5. No retest shall be permitted on a lot if it fails in the workability tests.

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4.3.4 Packaging, packing and marking.- Inspection for packaging, packing and marking shall be as directed by the contracting officer.

4.3.5 Inspection equipment.- The examinations shall be made using commercial measuring equipment with a precision of 10 percent of the tolerance on the product dimension.

4.4 Test method and procedures.

4.4.1 Hardness.- Each cup of the sample for this test shall be sectioned longitudinally. The specimen shall then be mounted on a flat anvil with the concave surface upward and readings taken near the center of the sidewall and tested in accordance with Method 243.1 in Federal Test Methods Standard No. 151.

4.4.2 Workability.- The work test shall comprise all of the process and production operations used in the manufacture of the cartridge case. (See 6.1.5)

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C (Domestic shipment).- The cups shall be packed in pallet boxes conforming to Drawings D7553671 and F7553844. Gross weight of each type box, with cups, shall not exceed 3000 pounds. Only one type of box shall be used for any one lot of cups.

5.2 Marking.- Each box shall be marked with the name of the material, lot number, of contractor and the contract or order number in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Drawing number of cup.

6.1.3 Type of packing.

6.1.4 Provision for submission of acceptance inspection reports, in duplicate, to the contracting officer, containing final inspection results for each lot of cups presented to the Government.

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6.1.5 Provision for performance of work test.- Where feasible, the work test portion of the initial production sample should be forwarded to the facility designated to receive the procured cups. That facility should perform the work test using tools, drawing and cleaning compounds, heat treating processes, lubricants and equipment, in accordance with the facility's manufacturing process. The remainder of the initial production sample should be forwarded to a Government approved facility for inspection. The work test on the initial production sample should not be performed until acceptability of the cups has been determined in the inspection phase and instructions have been issued by the Government to proceed with the test. In those instances where cups are being procured for delivery to multiple consignees, samples for work tests shall be furnished to each of the facilities.

6.2 The supplier shall certify that each lot of cups is protected by an adequate and readily removable rust preventive to insure their receipt in a satisfactory condition.

Custodian:

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Preparing activity:

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Project No. 1305-0535

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 REV MAR

PREVIOUS EDITION IS OBSOLETE.

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